Recognizing presentations by normal supported kernels

Human Compression Project

11 October 2026

Abstract

The construction of a compatible product and the recognition of a proposed presentation are different tasks. We give an order-free recognition criterion: normal local kernels, separately detected by coordinates, account for the whole kernel of a base map. This proves an actual isomorphism without first proving that the candidate group is finite. It gives a common proof of the presentation steps in the existing three-generator hypertope and square-section constructions. The local normality calculations remain explicit. The general injectivity criterion, direct-product isomorphism and quotient transport have been checked in Lean; the source-specific presentation and geometric hypotheses have written proofs here.

1 The recognition criterion

All homomorphisms in this paper are group homomorphisms. A product of subgroups means a product of sets until it has been shown to be a subgroup. No group is assumed finite in this section.

Theorem 1 (Normal kernel recognition)**.** *Let $\pi : P \to B$ and $\rho_i : P \to E_i$, for i in a finite set J, be homomorphisms. Suppose $K_i \trianglelefteq P$ satisfy*

$$\ker \pi = \langle K_i : i \in J \rangle, \tag{1}$$

$$\rho_i(K_j) = 1 \quad (i \neq j), \tag{2}$$

$$\ker \rho_i \cap K_i = 1. \tag{3}$$

Then the homomorphism

$$P \longrightarrow B \times \prod_{i \in J} E_i, \qquad x \longmapsto (\pi(x), (\rho_i(x))_i)$$

is injective. Moreover, the kernel in (1) *is the internal direct product of the K_i. For injectivity alone it suffices to replace* (1) *by* $\ker \pi \leq \langle K_i : i \in J \rangle$.

Proof. For $i \neq j$, an element of $K_i \cap K_j$ is killed by ρ_i by (2), and is therefore trivial by (3). Normality gives $[K_i, K_j] \leq K_i \cap K_j$, so distinct factors commute. Their finite set product is consequently their generated subgroup. If $\prod_i k_i = 1$ with $k_i \in K_i$, applying ρ_j gives $\rho_j(k_j) = 1$, whence $k_j = 1$. Thus multiplication identifies the external product with the generated subgroup. Finally, an element in the kernel of the displayed homomorphism belongs to $\ker \pi$, hence has such a product expression. Every coordinate vanishes, so every factor is trivial. □

Corollary 2 (Recognition of a quotient)**.** *Let $f : P \twoheadrightarrow H$ be a homomorphism, and let H have a base homomorphism $p : H \to B$ and coordinates $c_i : H \to E_i$. If Theorem 1 applies to pf, $c_i f$ and normal subgroups $K_i \leq P$, then f is an isomorphism.*

Proof. The coordinate map factors through f, so injectivity of the coordinate map implies injectivity of f. Surjectivity is the stated hypothesis. □

Normality is substantial. Let $P = D_8 = \langle u, v \mid u^2 = v^2 = (uv)^4 = 1\rangle$, let $B = 1$, and take $K_1 = \langle u\rangle$, $K_2 = \langle v\rangle$. Its abelianization $P \to C_2^2$ has coordinates faithfully detecting the two factors and killing the other one. The factors generate P, but the map has the nontrivial kernel $\langle (uv)^2\rangle$. Here neither factor is normal. Thus a correct local model and coordinate detection do not themselves eliminate mixed relations.

2 Supported extensions and complete presentations

We use the already proved supported-kernel theorem [UA, UA.SK], with the following full interface. Let $B = \langle b_t \,:\, t \in I\rangle$ have the intersection property. For each member S of a finite antichain $\mathcal{S}$ of subsets of I, there is a deletion $p_S : B \to B_S$, fixing the S-markers and killing the others, and a marked epimorphism $q_S : E_S \to B_S$ that is injective on every proper marked subgroup of E_S. Set $N_S = \ker q_S$. The compatible product

$$H = \{(b, (e_S)_S) \,:\, q_S(e_S) = p_S(b) \text{ for every } S\} \le B \times \prod_S E_S \tag{4}$$

is generated by its prescribed marker tuples. Its base kernel is $\prod_S N_S$, and its marked subgroup on $L \subseteq I$ has kernel $\prod_{S\subseteq L} N_S$. Its marking has the intersection property. The antichain assumption makes each $H_S \to E_S$ a marked isomorphism. These statements include generation of the *whole* group in (4); they are not assumptions that a subgroup has the ambient product's order.

Theorem 3 (Presentation recognition for an antichain)**.** *In the preceding setting, let $P = \langle r_t \,:\, t \in I\rangle$ be given by a presentation. Suppose:*

1. *its relators hold on the marker tuples in* (4)*;*
2. *the local relations of every E_S hold on $(r_t)_{t\in S}$;*
3. *the image K_S of N_S under $E_S \to P$ is normal in P;*
4. *adjoining the relations $K_S = 1$ for all S gives exactly the marked presentation of B.*

Then the marker map $P \to H$ is an isomorphism. In particular the K_S form an internal direct product, all marked intersections are exact, and if the input groups are finite then

$$|P_L| = |B_L| \prod_{S\subseteq L} |N_S|. \tag{5}$$

The additional deletion conclusion of UA.SK.Delete transfers to P under its hypotheses. The same holds for the geometric conclusion of UA.SK.Geometry under its hypotheses.

Proof. The first assumption and the generation assertion of UA.SK give a marked surjection $f : P \twoheadrightarrow H$. The second gives $E_S \to P_S$; its composite with f and projection to E_S is the identity on every local marker, hence is the identity homomorphism. Thus each local map is injective, without an order comparison. Under f, the image of N_S is its pure S-coordinate: the antichain makes every other intersection of supports proper, where q_T is injective. Consequently the coordinate maps kill K_S outside its own coordinate and are injective on K_S in that coordinate.

Normality makes $K = \langle K_S \,:\, S \in \mathcal{S}\rangle$ a normal subgroup. The last assumption identifies the marked map $P/K \to B$ with an isomorphism, so the base map $P \to B$ has kernel exactly K. Corollary 2 now proves that f is an isomorphism. The remaining conclusions follow from UA.SK and the explicitly indicated additional clauses. □

The last assumption is a presentation calculation. It does not assume the isomorphism $P \cong H$ which is being proved. In practice it says that, after killing the displayed local kernel words, the remaining relations are the known base presentation. A pair of marked maps which are inverse on generators also proves this condition. Normality likewise has a concrete local test: the markers within S already normalize K_S, and one must check the remaining markers. It is enough that those remaining markers centralize a generating set of K_S. Merely writing down the desired quotient does not supply this check.

3 The three-generator hypertope presentation

We retain the proved affine square model [UA, UA.Affine, UA.Square and UA.SquareDecomposition]. In coordinates, with the rightmost transformation acting first, it is

$$a(x,y) = (-x,y), \qquad b(x,y) = (y,x), \qquad c(x,y) = (x,1-y).$$

These generate $W = \mathbb{Z}^2 \rtimes D_8$ with complete presentation

$$\langle a,b,c \mid a^2 = b^2 = c^2 = (ab)^4 = (bc)^4 = (ac)^2 = 1\rangle. \tag{6}$$

For $\delta \geq 0$, define

$$Q_\delta = \begin{cases} W/q\mathbb{Z}^2, & \delta \text{ even}, \quad q = 2^{(\delta+2)/2}, \\ W/q\langle (1,1),(1,-1)\rangle, & \delta \text{ odd}, \quad q = 2^{(\delta+1)/2}. \end{cases}$$

It has order $2^{\delta+5}$, pair orders $(4,4,2)$ and all Boolean deletions. To identify the parity relation exactly, use $[x,y] = x^{-1}y^{-1}xy$. Direct affine multiplication gives translations

$$[(ab)^2, c] = (0,-2), \qquad [(ab)^2,(bc)^2] = (-2,-2).$$

The normal closure of a translation in W is the lattice spanned by its signed-permutation orbit. Hence, respectively, adding

$$[(ab)^2,c]^{2^{\delta/2}} = 1 \quad \text{or} \quad [(ab)^2,(bc)^2]^{2^{(\delta-1)/2}} = 1 \tag{7}$$

to (6) gives exactly Q_δ.

Take $n \geq 10$, $s,t \geq 2$, $\ell \geq 1$, and $\delta = n - s - t - \ell \geq 0$. Let P have the three involutory markers, the pair-power relations

$$(ab)^{2^s} = (bc)^{2^t} = (ac)^{2^\ell} = 1,$$

the three relations

$$[(ab)^4, c] = [a,(bc)^4] = [(ac)^2, b] = 1, \tag{8}$$

and the appropriate parity relation (7). This is the defining presentation of [HFL19, Theorem3.2], with renamed markers.

Corollary 4. *The presentation P is the supported extension of Q_δ by the three marked dihedral quotients of orders $2^{s+1}, 2^{t+1}, 2^{\ell+1}$ onto its pair subgroups. In particular it has order 2^n, the prescribed pair orders, and a regular rank-three hypertope with full type-preserving group P. The subgroups*

$$K_{ab} = \langle (ab)^4\rangle, \qquad K_{bc} = \langle (bc)^4\rangle, \qquad K_{ac} = \langle (ac)^2\rangle$$

form its direct-product base kernel, of type $C_{2^{s-2}} \times C_{2^{t-2}} \times C_{2^{\ell-1}}$.

Proof. The three pair supports are an antichain. Their proper marked subgroups are the trivial group and the reflection subgroups, so the maps to the base are injective on every proper marked subgroup. UA.SK therefore constructs H.

All relators of P hold in H. Each pair-power kernel word is supported only in its own coordinate, where the third marker is the identity, proving (8). The two commutator words in (7) vanish in each

local pair coordinate: one of the relevant markers is deleted, and direct substitution gives the identity. Their base values are the specified translations. Each local dihedral presentation holds in P. Within its pair, each marker inverts the displayed kernel generator; the third marker centralizes it by (8). Thus all three K's are normal. Killing them reduces the presentation precisely to (6) and (7). Theorem 3 applies.

Its order formula gives $2^{\delta+5}2^{s-2}2^{t-2}2^{\ell-1} = 2^n$ and its local recognition gives the exact pair orders. The square base is a regular string polytope; its maximal-parabolic cosets have the needed flag property by the retained polytope correspondence. The dihedral local markers are involutions, so UA.SK.Geometry supplies the claimed hypertope and its full group. This last use is explicit: intersection property alone does not supply the flag property of a general coset geometry. □

The quotient description is consequently

$$P \cong (C_{2^{s-2}} \times C_{2^{t-2}} \times C_{2^{\ell-1}}) \,.\, \begin{cases} (D_{2^{(\delta+4)/2}} \times D_{2^{(\delta+4)/2}}) \rtimes C_2, & \delta \text{ even}, \\ (D_{2^{(\delta+3)/2}} \times D_{2^{(\delta+3)/2}}) \rtimes C_2^2, & \delta \text{ odd}. \end{cases}$$

Here the extension dot makes no splitting assertion. These are the corrected parity rows already recorded in the accepted source contract; the source interchanges them and repeats the first pair word when naming its second kernel. The present argument preserves both corrections. The decompositions of the square quotient are the previously proved UA.SquareDecomposition, not new group classifications.

4 Square sections along a string

Let $d \geq 3$ and $n_3, \dots, n_d \geq 5$. Write $S_i = \{i, i+1, i+2\}$ for $0 \leq i \leq d-3$, and let $E_i = H_{n_{i+3}}$ be the square group of UA.Square. Its base quotient is H_5, and its kernel is $\gamma_3(E_i)$, generated by its two doubled coordinate translations. Those translations need not generate independent cyclic factors: in H_6 they coincide. Their full subgroup, rather than a presumed product of two cyclic orders, is used throughout.

The common base B is the class-two path group with d binary vertex coordinates and $d-1$ binary edge coordinates. Its multiplication is

$$(v, z)(w, t) = (v + w, z + t + \beta(v, w)), \qquad \beta_i(v, w) = v_{i+1}w_i.$$

The full coordinate model and presentation are proved in UA.Forest: its order is 2^{2d-1}, its marked subgroups retain exactly the internal vertex and edge coordinates, and it has every Boolean deletion. Each $B_{S_i} = H_5$. The quotient $E_i \to H_5$ is faithful on every proper marked subgroup, since all pair orders remain $4, 4, 2$. Thus UA.SK gives a group H with precisely these square sections and order

$$|H| = 2^{2d-1+\sum_{j=3}^{d}(n_j-5)} = 2^{\sum_{j=3}^{d} n_j - 3(d-3)}. \tag{9}$$

This also identifies H with the particular iterated mix constructed in [CFHS25, Theorem5.1], not merely with a group of the same order and local type. Indeed, the projection $H \to \prod_i E_i$ is injective. If every local coordinate is trivial, then every triple deletion of its base coordinate is trivial. Every vertex and edge of the path lies in a consecutive triple, so the displayed coordinates of B force that base element to be trivial. The projected image is generated by the marker tuples in the local factors. The source's definition (8) and its induction in Theorem5.1 produce exactly these tuples, with only a reassociation of the direct product. Hence they give the same marked group.

Let P_0 have the string Coxeter relations of type $\{4, \dots, 4\}$ and the complete local lattice relations of each E_i. Let P be its quotient by

$$[(r_i r_{i+1})^2, (r_{i+2} r_{i+3})^2] = 1 \qquad (0 \leq i \leq d-4). \tag{10}$$

Lemma 5 (The local normalization calculation)**.** *If involutions* a, b, c, d *satisfy* $[a, c] = [a, d] = [b, d] = 1$ *and* $[(ab)^2, (cd)^2] = 1$*, then* d *commutes with both* $(cbab)^2$ *and* $(abcb)^2$*.*

Proof. Put $x = bab$, $A = (ab)^2 = ax$, and $F = (cd)^2$. As a commutes with F, the relation $[A, F] = 1$ gives $[x, F] = 1$. Also $[x, d] = 1$, so $[x, Fd] = [x, cdc] = 1$. Conjugating by c gives $[cxc, d] = 1$. Both factors of $(cbab)^2 = (cxc)x$ commute with d. Finally $abcb = b(cbab)^{-1}b$, and b commutes with d. □

Corollary 6. *The marked homomorphism* $P \to H$ *is an isomorphism. Furthermore* P_0 *is the universal regular polytope group with the indicated consecutive square sections, and* P *is precisely its quotient by* (10)*. Thus the presentation assertion in [CFHS25, Theorem5.3] follows from Theorem 3.*

Proof. The complete local relations hold in H by local recognition. Relations (10) hold in its base, which is class two. In each triple coordinate one of the two disjoint edges loses an endpoint, and the square of its product is the identity. Thus all relations hold in every coordinate, giving $P \to H$. The local E_i also map into P by its defining relations.

Let K_i be the image of $\gamma_3(E_i)$. It is normalized by the three local markers. The two doubled translations are the words in Lemma 5; that lemma and its reversed version show that the nearest outside markers centralize them. All more distant markers commute with every local marker by the string relations. Hence $K_i \trianglelefteq P$.

After killing the K_i, each consecutive triple is class two. Every adjacent square therefore commutes with its endpoints and its nearest outside markers; it commutes with all remaining markers by the string relations. Thus the quotient is class two, with the defining involutions, distant commutations and adjacent product orders dividing four. It is a quotient of the complete class-two path presentation of B. Conversely the map onward to the base B is a marked surjection. These two maps are inverse on generators, so this quotient is exactly B. This proves the last condition of Theorem 3 without an upper bound on a finite order. The theorem now identifies $P = H$, with (9) following from the already established compatible-product formula.

For universality, every defining relator of P_0 survives all deletion assignments. This is immediate for even Coxeter relators and is the proved lattice-relator calculation in UA.Square for local relations. Hence P_0 has Boolean deletions and the intersection property by UA.D, without an assumption of finiteness. Its local square subgroups embed because their maps onward to H are faithful; its markers remain nonidentity involutions. It is therefore a string C-group. Every marked C-group with the prescribed local sections satisfies its defining relations, and is its marked quotient. The retained regular-polytope correspondence gives precisely the universal polytope assertion. No order of P_0 is asserted. □

5 What has been shared and what remains local

The two applications used to prove separate upper bounds on a candidate presentation and compare them to a constructed finite quotient. Their shared replacement is now Theorem 3: it proves faithfulness before any order is calculated, and also works for infinite local groups. The underlying idea grows from the already reviewed UA.SK and UA.Order arguments. We make no historical priority claim for the resulting abstract group lemma. The geometric results retain their original attribution.

The genuine source-specific work is visible: in the hypertope case it is (7), the three normality relations and the square quotient; in the string case it is Lemma 5 and the class-two quotient identification. Neither is removed by renaming it an umbrella input. The supported-kernel construction, affine square models, and appropriate geometric correspondence are still required dependencies. This note does not replace the bounded cyclic-core classification in L035, the finite census and presentation checks in N009, or all of F016. No source or theorem credit and no compression ratio is assigned.

The Lean file `NormalKernelRecognition.lean` proves a finite-join coordinate lemma, the joint injectivity assertion in Theorem 1, and Corollary 2, in arbitrary groups. It derives the required

product decomposition from subgroup normality; it does not take final injectivity as a premise. The companion file `NormalKernelProduct.lean` constructs the explicit multiplicative isomorphism $\prod_i K_i \simeq \ker \pi$. Its map is multiplication in P, and on each individual factor it is the given subgroup inclusion. This last conclusion uses equality in (1); containment alone would not imply it. Fourteen declaration reports use only the standard Lean axioms. The formal construction uses mathlib's existing multiplication map for commuting independent subgroups, also used in earlier project components.

These files do not formalize the source presentations, the construction of UA.SK, the normality checks in the applications, the affine calculations or the geometric correspondence in this note. Those remain the written obligations displayed above. An existing Lean lemma, `BooleanRetractions.four_marker_centralization`, verifies Lemma 5; applying it to all local kernels still requires the written argument in the final application.

References

[1] [UA] Human Compression Project, *Supported extensions and finite two-group quotients*, current Topic A umbrella; reviewed phase207 proofs, canonical TeX edition. Required results are UA.D, UA.SK, its local, deletion and geometric clauses, UA.Forest, UA.Affine, UA.Square and UA.SquareDecomposition.

[2] [HFL19] D.-D. Hou, Y.-Q. Feng and D. Leemans, *Existence of regular 3-hypertopes with* 2^n *chambers*, arXiv:1901.09034v1, 23 January 2019, Theorem3.2.

[3] [CFHS25] G. Cunningham, Y.-Q. Feng, D.-D. Hou and E. Schulte, *Symmetric polytopes whose automorphism groups are 2-groups*, arXiv:2512.15511v1, 18 December 2025, Theorems5.1 and5.3.

[4] [MS] P. McMullen and E. Schulte, *Abstract Regular Polytopes*, Cambridge University Press, 2002, Section2E; the correspondence is retained through the precise UA interface.